

LEVEL 5 · WRITE-UP

Parking Orbit

Step 1 — orbit radius

Plug the given constants into $r = (G \cdot M \cdot T^2 / 4\pi^2)^{(1/3)}$:

$$\begin{aligned} G \cdot M \cdot T^2 &= 6.674e-11 \cdot 5.972e24 \cdot 86164^2 \\ &= 2.9596e24 \\ \div 4\pi^2 &= 7.4960e22 \\ r &= (7.4960e22)^{(1/3)} \approx 4.2163e7 \text{ m} = 42163 \text{ km} \end{aligned}$$

Step 2 — altitude (the trap)

The seed is the altitude, i.e. the radius minus Earth's radius. Solvers who stop at the orbit radius (42163) will decode only noise.

$$\text{altitude} = 42163 \text{ km} - 6371 \text{ km} = 35792 \text{ km} \quad \leftarrow \text{this is the seed}$$

Step 3 — run the lock

Set SEED = 35792 in lock.py and run it. The key-stretch + SHA-256 keystream XORs against the ciphertext and prints the phrase:

```
import base64, hashlib
SEED = 35792
ITER = 1 << 20
def stretch(s):
    h = str(s).encode()
    for _ in range(ITER): h = hashlib.sha256(h).digest()
    return h
def keystream(s, n):
    base = stretch(s); out=b''; c=0
    while len(out) < n:
        out += hashlib.sha256(base + c.to_bytes(4,'big')).digest(); c+=1
    return out[:n]
ct = base64.b64decode('WDEE0t3A10S44tHJ3QK0MxmBIjJzCw==')
print(bytes(a^b for a,b in zip(ct, keystream(SEED, len(ct)))).decode())
# -> What_A_Wonderful_World
```

CTC{What_A_Wonderful_World}

Why there is no shortcut

- SHA-256 is one-way: you cannot run the keystream backwards from ciphertext to flag.

- The 2^{20} -round stretch makes brute-forcing the seed space expensive, so looping over every 5-digit number until text appears is impractical.
- The keystream is indistinguishable from random, so the plaintext cannot be read off by eye or inferred by a language model — the correct altitude must actually be computed and supplied.

Builder notes

- Ship both files: `transmission.txt` and `lock.py`.
- Flag is exact-match; set it case-sensitive since the phrase uses specific capitals.
- Cheap hint: “Radius is not altitude.” Dearer hint: “seed = 35792.”
- Self-contained physics/code level; its only chain role is to unlock Level 6.